Quantification: 535_z13c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	12
interior lacuna count	12
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 12)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	2998.92	3028.5	2648.47	px^2
major_axis_length_px	lacuna shape	86.7847	96.5833	46.8513	px
minor_axis_length_px	lacuna shape	42.1183	41.2248	19.3946	px
aspect_ratio	lacuna shape	1.9935	2.0511	0.5058	unitless
eccentricity	lacuna shape	0.821	0.873	0.1246	unitless
solidity	lacuna shape	0.8195	0.8055	0.0847	unitless
roots_count	headline	8.6667	8.5	5.5158	count
ring_length_r30_px	headline	297.667	289	177.281	px
ring_length_r60_px	reported	577.75	485.5	334.91	px
owned_length_px	ownership-dependent	757.061	584.881	652.073	px
edge_count	ownership-dependent	36.75	26.5	33.912	count
mean_edge_length_px	ownership-dependent	23.3821	20.6761	10.5563	px
perimeter_px	normalised	274.172	267.472	181.324	px
ring_area_r30_px2	normalised	9040.83	10217	4294.25	px^2
ring_area_r60_px2	normalised	19911.7	22575.5	7615.23	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.999	1	0.0033	unitless
ring_density_r30	normalised	0.0316052	0.0333819	0.00846137	px^-1
ring_density_r60	normalised	0.0280303	0.0266268	0.0087932	px^-1
roots_per_100px_perimeter	normalised	3.3169	2.9823	1.68	per 100 px
ring_attached_length_r30_px	network v2	267.5	249	178.851	px
ring_attached_length_r60_px	network v2	460.917	387.5	362.307	px
ring_length_w_r30_px	network v2	336.203	326.001	203.272	px
ring_length_w_r60_px	network v2	654.226	535.197	384.387	px
sholl_crossings_r10	network v2	7.3333	7	4.4992	count
sholl_crossings_r20	network v2	7.5	9	4.5025	count
sholl_crossings_r30	network v2	8.25	7.5	4.9383	count
loop_count	loops, report only	1.5833	0	2.2747	count
loop_edge_share	loops, report only	0.0761	0	0.0994	share
ring_width_mean_r30_px	width	7.1542	7.3194	0.826	px

Per field
----

measure	value	units
analysed_area_px2	1.01259e+06	px^2
total_skeleton_length_px	21999	px
canalicular_length_density_per_px	0.0217255	px^-1
junction_count	391	count
junction_density_per_px2	0.00038614	px^-2
lacuna_count	12	count

lacunae_per_px2	1.185e-05	px^-2
skeleton_component_count	347	count
mean_component_length_px	63.3977	px
median_component_length_px	19	px
field_length_density_w_per_px	0.0244391	px^-1
field_density_without_flagged_per_px	0.0212729	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	725	px
unattached_component_count	44	count
canalicular_length_density_attached_per_px	0.0217255	px^-1
field_length_density_with_unattached_per_px	0.0224415	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	30	count
loop_edge_share	0.0882	share
edge_count	1123	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.1904	px
width_median_px	7.2111	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	20443	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	3449	94.2445	51.4301	1.8325	0.838	0.8008	-1.3442	87.5868	746.098	False
2	916	42.5998	32.8395	1.2972	0.637	0.7829	0.4554	434.044	377.415	False
3	6729	150.978	65.1171	2.3186	0.9022	0.7721	-0.3046	496.001	291.95	False
4	3320	105.162	44.1209	2.3835	0.9077	0.7721	-1.4464	464.635	577.918	False
5	3645	102.737	49.6538	2.0691	0.8754	0.8697	0.3597	580.157	513.438	False
6	2801	98.9222	38.3287	2.5809	0.9219	0.8173	0.524	580.027	151.776	False
7	9054	174.453	85.8111	2.033	0.8707	0.6586	0.1605	649.475	679.118	False
8	1209	70.3083	25.9548	2.7089	0.9294	0.8103	-1	737.055	638.021	False
9	3256	105.273	45.222	2.3279	0.903	0.7485	-1.2191	649.232	325.526	False
10	543	31.4075	23.6295	1.3292	0.6588	0.9282	0.087	780.65	298.357	False
11	455	27.4749	22.5328	1.2193	0.5722	0.9362	1.1384	784.457	318.629	False
12	610	37.8566	20.7792	1.8218	0.8359	0.937	-0.7458	884.313	871.107	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	9	457	1078	1649.21	86	19.1768	7.03
2	10	250	450	635.073	35	18.1449	7.2607
3	18	521	982	1575.45	86	18.3192	7.7038
4	8	335	501	568.304	24	23.6793	6.4846
5	4	255	470	530.304	16	33.144	7.7753

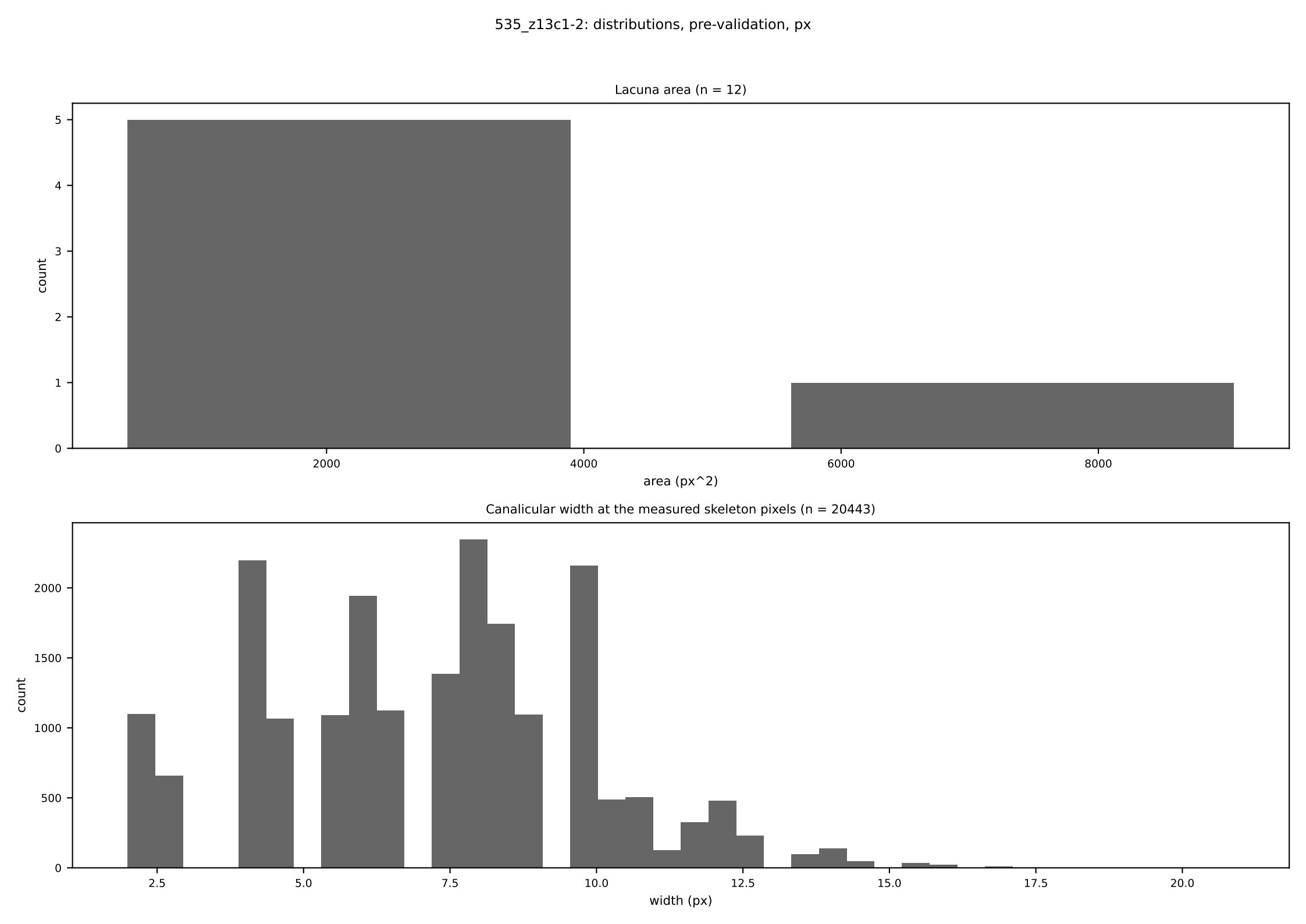
6	12	323	618	601.458	31	19.4019	7.3781
7	15	592	1170	2092.86	99	21.14	7.7763
8	6	94	141	211.995	4	52.9987	5.8203
9	15	441	617	634.399	29	21.8758	7.4882
10	2	101	349	164.811	8	20.6014	8.3628
11	4	98	207	352.764	17	20.7509	7.2126
12	1	105	350	68.1127	6	11.3521	5.5571

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	311.848	10999	27029	1	0.9885	0.0415492	0.0398831	2.886
2	145.782	5773	13561	1	1	0.043305	0.0331834	6.8596
3	433.103	13694	24741	1	1	0.0380459	0.0396912	4.1561
4	326.563	11164	23645	1	1	0.0300072	0.0211884	2.4498
5	270.936	10229	21506	1	1	0.0249291	0.0218544	1.4764
6	264.007	10205	23754	1	1	0.0316512	0.0260167	4.5453
7	731.429	16860	33118	1	1	0.0351127	0.0353282	2.0508
8	194.894	5462	13139	1	1	0.0172098	0.0107314	3.0786
9	328.877	11820	23768	1	1	0.0373096	0.0259593	4.561
10	96.0833	3803	10045	1	1	0.026558	0.0347437	2.0815
11	85.734	2789	7600	1	1	0.035138	0.0272368	4.6656
12	100.811	5692	17034	1	1	0.0184469	0.0205471	0.992

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	449	932	511.718	1232.23	11	10	12
2	239	402	277.995	511.446	7	6	7
3	452	920	593.872	1118.35	13	12	16
4	256	304	369.35	551.475	7	13	6
5	242	373	282.652	518.919	8	9	5
6	306	495	375.433	701.612	7	10	9
7	568	1123	672.742	1333.1	15	13	16
8	94	120	107.326	162.853	2	2	1
9	433	572	510.659	706.269	11	9	12
10	78	168	112.426	399.049	2	1	8
11	61	90	105.012	224.024	2	3	2
12	32	32	115.255	391.392	3	2	5


535_z13c1-2: distributions, pre-validation, px
Lacuna area (n = 12)
count
area (px^2)
0
1
2
3
4
5
2000
4000
6000
8000
Canalicular width at the measured skeleton pixels (n = 20443)
count
width (px)
0
500
1000
1500
2000
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0